UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS

Paper 4 Alternative to Practical Test

5054/04

October/November 2006

1 hour

Candidates answer on the Question Paper.
No Additional Materials are required

Candidate Name

Centre Number

Candidate Number

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **9** printed pages and **3** blank pages.

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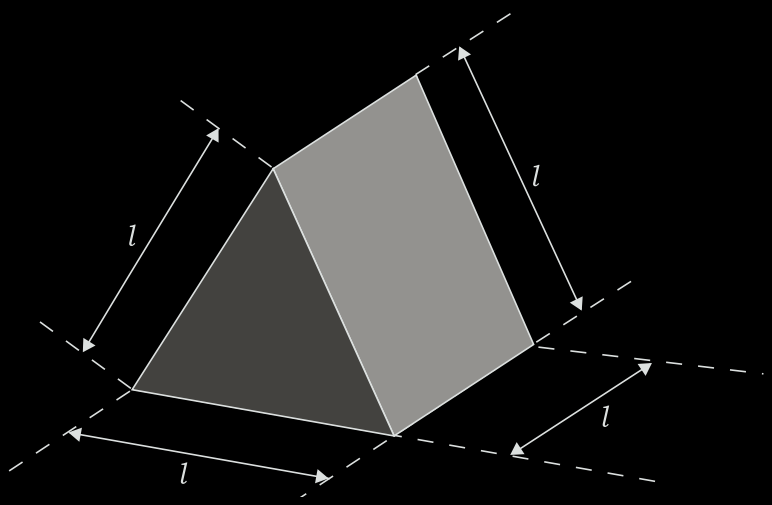


Fig. 1.1

The volumes measured were

24.8 cm^3 27.8 cm^3 24.5 cm^3 24.3 cm^3 25.0 cm^3.

(a) **(i)** Circle the measurement that is **not** consistent with the others. [1]

(ii) Suggest a possible reason for the error in this measurement.

..... [1]

(iii) Calculate the average value for V, ignoring the inconsistent measurement.

V = ... [2]

(b) The volume of the prism is given by

$$V = 0.433 l^3$$

where l is the length of the side of the prism.

Use your average value for V to obtain a value for l. Give your answer to a suitable number of significant figures.

l = ... [2]

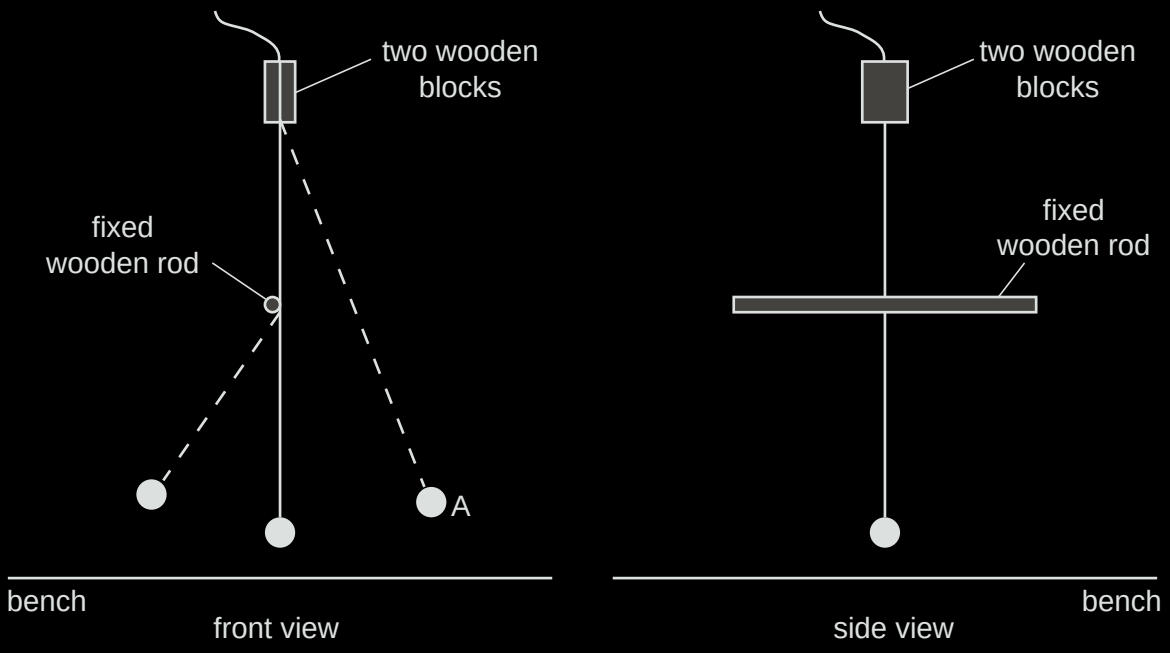


Fig. 2.1

In the experiment, the height h of the centre of the wooden rod above the bench is varied. The time t for one complete oscillation is obtained for each value of h.

(a) On Fig. 2.1, mark accurately

(i) the height h, [1]

(ii) where the student's eye should be positioned when measuring t. [1]

(b) **(i)** Describe how the student could ensure that the wooden rod is horizontal. You may draw on the diagram if you wish.

.....

..... [1]

(ii) Suggest why the wooden rod should be horizontal.

.....

..... [1]

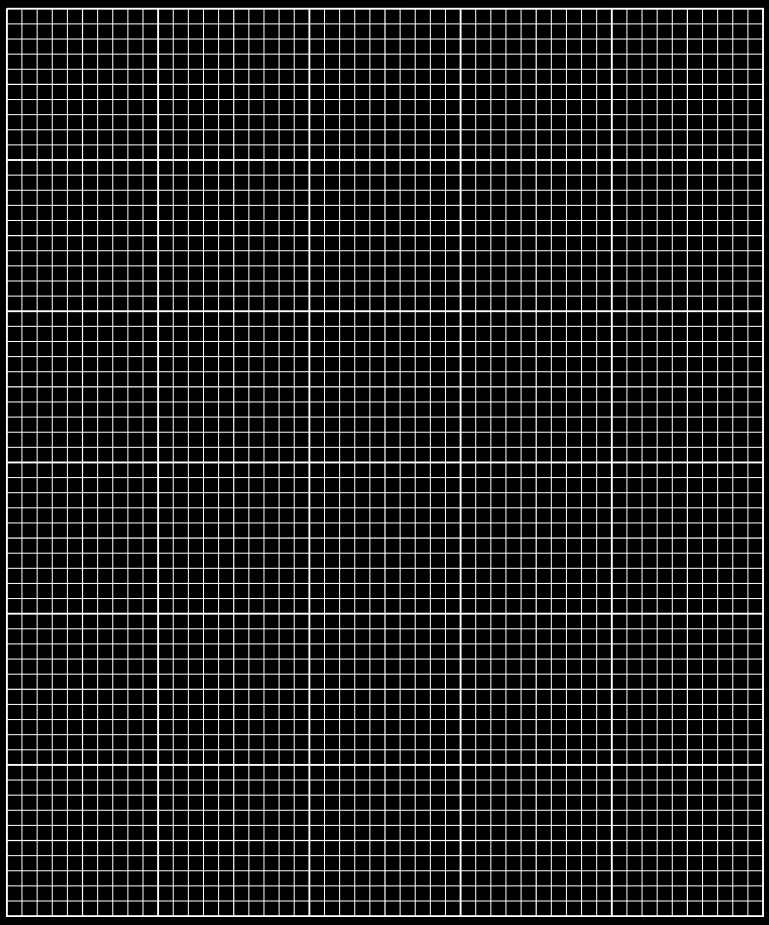
(c) The value of t is approximately 1 s. Describe how the student could obtain precise values for t.

.....

..... [1]

Question 2 continues on page 4

h / cm	t / s
40	1.26
35	1.22
30	1.18
25	1.12
20	1.05
15	0.98
10	0.90



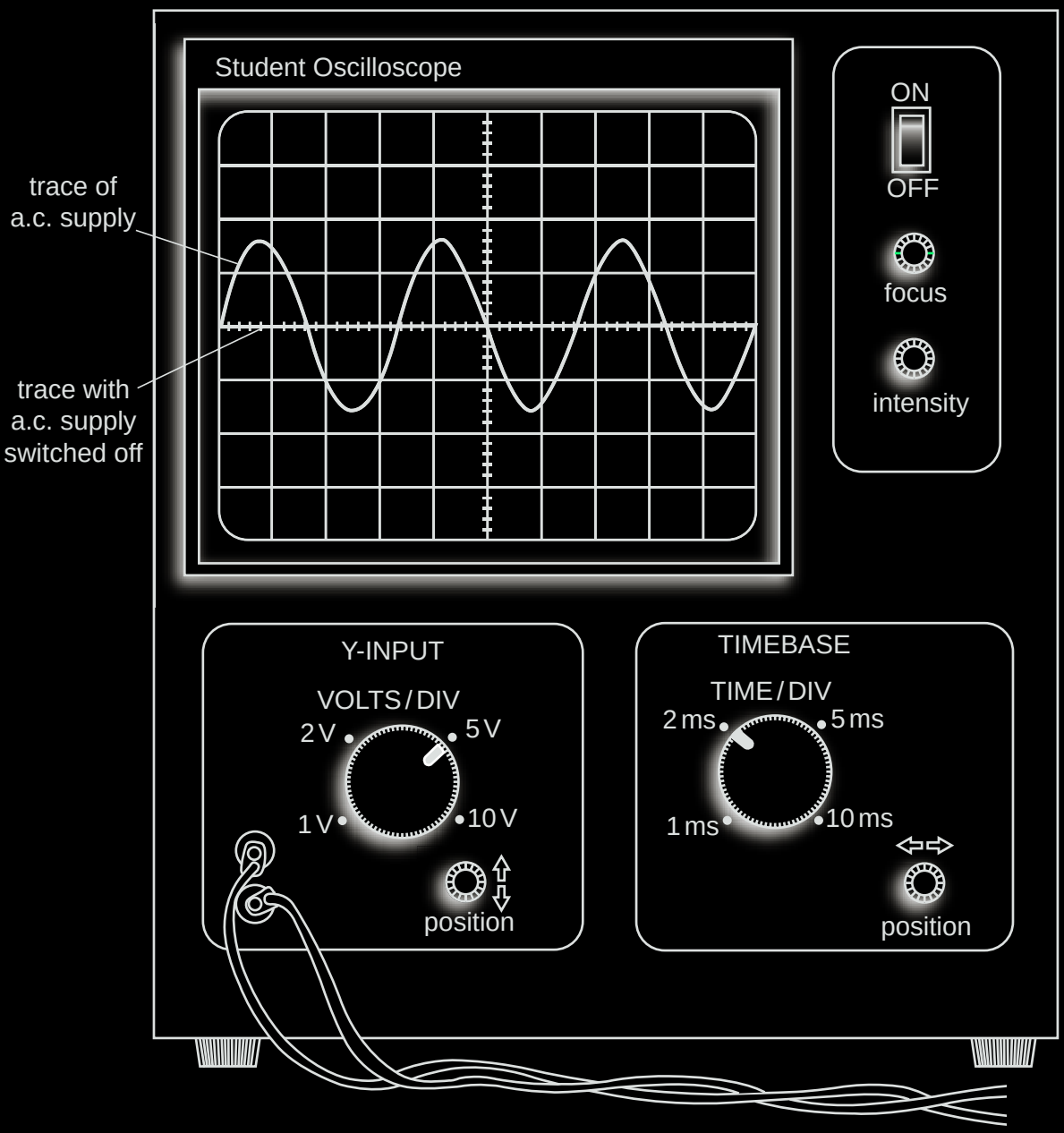

Student Oscilloscope
trace of a.c. supply
trace with a.c. supply switched off
ON
OFF
focus
intensity
Y-INPUT
VOLTS / DIV
2 V
5 V
1 V
10 V
position
TIMEBASE
TIME / DIV
2 ms
5 ms
1 ms
10 ms
position

Fig. 3.1

$f = \frac{1}{T}$ to find the frequency f of the a.c. supply.

f = ... [1]

(c) **(i)** Explain why it would **not** be possible to measure the frequency of an a.c. supply of frequency 15 Hz using the CRO on these settings.

.....

..... [1]

(ii) Suggest which setting for the time-base could be used when measuring a frequency of 15 Hz.

..... [1]

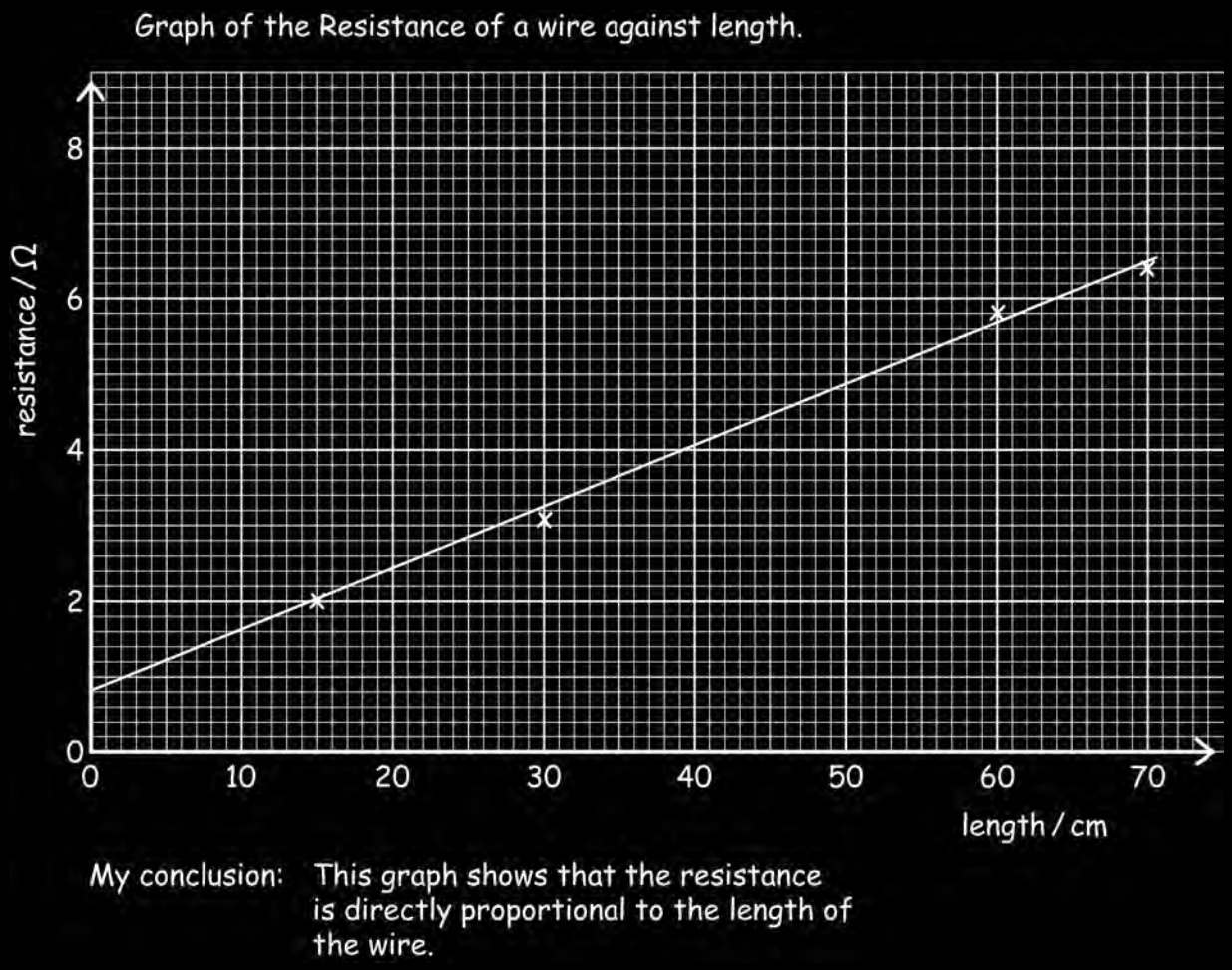


Fig. 4.1

(a) By taking readings from the graph, draw a table showing the results the student obtained.

[3]